

ISyE6669-OAN Homework Week 9

Fall 2025

1. Directly write down the dual linear programs of the following linear programs. You do not need to write down the Lagrangian relaxation procedure.

(a)

$$\begin{array}{ll}\max & x + y + z \\ \text{s.t.} & 10x - y + 5z \leq 2, \\ & 2y - 3z = 3, \\ & x + y - 2z \geq -4, \\ & x \geq 0, y \leq 0, z \text{ free.}\end{array}$$

Solution:

$$\begin{array}{ll}\min & 2y_1 + 3y_2 - 4y_3 \\ \text{s.t.} & 10y_1 + y_3 \geq 1, \\ & -y_1 + 2y_2 + y_3 \leq 1, \\ & 5y_1 - 3y_2 - 2y_3 = 1, \\ & y_1 \geq 0, y_2 \text{ free}, y_3 \leq 0.\end{array}$$

(b)

$$\begin{array}{ll}\min & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} \leq \mathbf{b}, \\ & \mathbf{x} \geq \mathbf{0}.\end{array}$$

Solution:

$$\begin{array}{ll}\max & \mathbf{b}^\top \mathbf{y} \\ \text{s.t.} & \mathbf{y}^\top \mathbf{A} \leq \mathbf{c}^\top, \\ & \mathbf{y} \leq \mathbf{0}.\end{array}$$

2. Long John Jarvis discovered a vast treasure hidden on a deserted island. There was an unlimited amount of gold and silver coins, jewel-encrusted plates, and ivory statues, all of which were completely unguarded and could be taken at no risk. Unfortunately, Long Jong's ship had been sunk, and he only had a small rowboat to put his treasure. The rowboat could hold up to 400 pounds of treasure with a volume up to 500 cubic feet. The following table gives the value, size, and weight of each type of treasure:

	Unit Value (thousands of dollars)	Size (cubic feet)	Weight (pounds)
Gold coin	4	2	3
Silver coin	1	1	2
Jewel-encrusted plate	3	3	1
Ivory Statue	10	4	6

Being a very worldly and intelligent pirate, Long John was familiar with the basic principles of linear programming. He came up with the following decision variables x_1, x_2, x_3, x_4 to be the numbers of gold coins, silver coins, jewel-encrusted plates, and ivory statues that he would take. Then, he set up the following LP to decide how much each treasure he should take in order to maximize his wealth.

$$\begin{aligned}
 (P) \quad & \max \quad 4x_1 + x_2 + 3x_3 + 10x_4 \\
 & \text{s.t.} \quad 2x_1 + x_2 + 3x_3 + 4x_4 \leq 500 \\
 & \quad \quad 3x_1 + 2x_2 + x_3 + 6x_4 \leq 400 \\
 & \quad \quad x_1, x_2, x_3, x_4 \geq 0.
 \end{aligned}$$

Because the rowboat's limits were merely estimations, Long John was not concerned with restricting his variables to be integers; he could just round up the solution to the nearest integers if necessary.

A much more important problem for Long John was that his laptop with python and gurobi was destroyed when his ship was sunk, so he could only solve LPs graphically by drawing on the sand with a stick. However, since his LP had four variables, it would require a 4-dimensional graph to solve, which was obviously impossible for Long John to draw.

Not knowing how to solve his LP, Long John just grabbed as many ivory statues (66 statues) as he could and returned home with a wealth of \$660K. However, as a more advanced student of linear programming than Long John was, you can do better! Work through the following steps to retrieve more treasure!

- (a) Form the dual minimization problem of the above LP that Long John came up with.

Solution:

$$\begin{aligned}
 \min \quad & 500y_1 + 400y_2 \\
 \text{s.t.} \quad & 2y_1 + 3y_2 \geq 4, \\
 & y_1 + 2y_2 \geq 1, \\
 & 3y_1 + y_2 \geq 3, \\
 & 4y_1 + 6y_2 \geq 10, \\
 & y_1 \geq 0, y_2 \geq 0
 \end{aligned}$$

- (b) Use a computer program to draw the feasible region of your dual problem. Hopefully you don't need a 4-dimensional graph!

Solution:

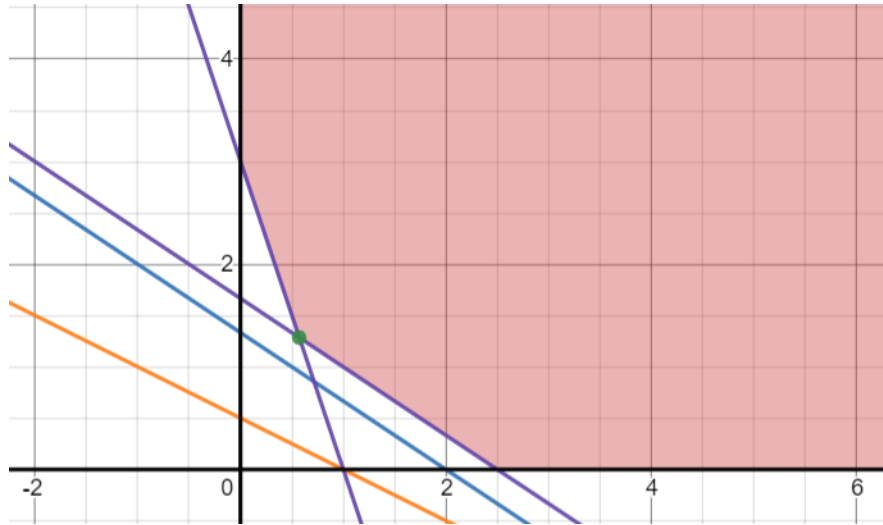


Figure 1: Feasible region of the dual problem

- (c) Point out on the graph the optimal solution of the dual problem. Find out the numerical values of this optimal dual solution. Hint: You might need to solve a simple set of linear equations, which you can do by writing on the sand, but no need to run any simplex method.

Solution: The optimal solution is $y_1^* = \frac{4}{7}$, $y_2^* = \frac{9}{7}$ at the intersection of $3y_1 + y_2 = 3$ and $4y_1 + 6y_2 = 10$. It is indicated by the green point in the above graph.

- (d) What is the optimal objective value of the dual problem? So, how much more wealth can you get than Long John did? Notice that we haven't figured out the primal optimal solution yet. So continue.

Solution: The minimum objective value of the dual problem is 800. This is an increase of \$140k from the original amount (\$660k).

- (e) By just looking at the dual problem, decide which primal constraints must be active at the primal optimal solution. Hint: Look at which dual optimal variables are nonzero. Then use the Primal Complementary Slackness.

Solution: By complementary slackness,

$$\begin{aligned} y_1^* = \frac{4}{7} > 0 &\implies 2x_1 + x_2 + 3x_3 + 4x_4 = 500 \\ y_2^* = \frac{9}{7} > 0 &\implies 3x_1 + 2x_2 + x_3 + 6x_4 = 400 \end{aligned}$$

So both primal constraints are active at the primal optimal solution.

- (f) By just looking at the dual problem, decide which primal variables of the optimal primal solution must be zero. Hint: Look at which dual constraints are active, and which are not active at the dual optimal solution. Then, use the Dual Complementary Slackness.

Solution: Plugging in $y_1^* = \frac{4}{7}, y_2^* = \frac{9}{7}$,

$$2y_1 + 3y_2 - 4 = 1 > 0 \implies x_1^* = 0$$

$$y_1 + 2y_2 - 1 = \frac{15}{7} > 0 \implies x_2^* = 0$$

$$3y_1 + y_2 - 3 = 0 \implies x_3^* \neq 0$$

$$4y_1 + 6y_2 - 10 = 0 \implies x_4^* \neq 0$$

So primal variables x_1^* and x_2^* are zero at the primal optimal solution.

- (g) Now, find out the primal optimal solution. This again you can do by just writing on the sand. You can grab the treasure now!

Solution: Since we found that x_1^* and x_2^* are zero and both constraints are active at the optimal solution, we get the following set of linear equations:

$$3x_3 + 4x_4 = 500$$

$$x_3 + 6x_4 = 400$$

We solve them to get the optimal primal solution:

$$x^* = \begin{bmatrix} 0 \\ 0 \\ 100 \\ 50 \end{bmatrix},$$

and we can go and grab 100 jewel-encrusted plates and 50 ivory statues.

3. Let A be a $n \times n$ skew-symmetric matrix. Remember that a skew-symmetric matrix is a matrix which is equal to the negative of its transpose, i.e., $A = -A^\top$. (For example: $\begin{bmatrix} 0 & 2 & -7 \\ -2 & 0 & -1 \\ 7 & 1 & 0 \end{bmatrix}$ is a 3×3 skew-symmetric matrix.) Consider the linear program:

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq -c \\ & x \geq 0. \end{aligned} \tag{1}$$

We will assume that the above problem is feasible and bounded.

- (a) Write down the dual LP of the above LP (1).

Solution: Dual of the problem (1) is:

$$\begin{aligned} \min \quad & -c^\top y \\ \text{s.t.} \quad & y^\top A \geq c^\top \\ & y \geq 0. \end{aligned} \tag{2}$$

- (b) Show that the feasible region of the primal LP and the dual LP is exactly the same, that is show that (a) if x^* is a feasible solution of the primal LP, it is also a solution of the dual LP, and (b) if y^* is a solution of the dual LP it is also a solution of the primal LP. [Note: This is a very rare property which is occurring due to the special-structure of the above LP. Typical LPs will never have this property.]

Solution: From

$$y^\top A \geq c^\top \Leftrightarrow A^\top y \geq c \Leftrightarrow -A^\top y \leq -c$$

and the fact $-A^\top = A$ we get:

$$y^\top A \geq c^\top \Leftrightarrow Ay \leq -c.$$

Hence we can write dual as:

$$\begin{aligned} \min \quad & -c^\top y \\ \text{s.t.} \quad & Ay \leq -c \\ & y \geq 0. \end{aligned} \tag{3}$$

- (c) Show that the optimal objective function value of this problem is 0. [Hint: use result of above question, structure of the objective function of the dual, and strong duality result.]

Solution: As we seen in part (b), both primal and dual have same feasible set: $P = \{x \in \mathbb{R}^n : Ax \leq -c, x \geq 0\} = \{y \in \mathbb{R}^n : Ay \leq -c, y \geq 0\}$. Lets denote optimal solution of primal as $x^* \in P$ and optimal solution of the dual as $y^* \in P$. From strong duality we have

$$c^\top x^* = -c^\top y^* = v_{opt} \tag{4}$$

where v_{opt} is optimal value of both primal and dual problem.

Since, x^* is an optimal solution of the maximization problem, it means that for every $p \in P$ we have $c^\top p \leq c^\top x^*$. By setting $p = y^*$ and using (4) we get:

$$-v_{opt} = c^\top y^* \leq c^\top x^* = -c^\top y^* = v_{opt} \Rightarrow v_{opt} \geq 0$$

Since, y^* is an optimal solution of the minimization problem, it means that for every $p \in P$ we have $-c^\top y^* \leq -c^\top p$. By setting $p = x^*$ and using (4) we get:

$$v_{opt} = -c^\top y^* \leq -c^\top x^* = c^\top y^* = -v_{opt} \Rightarrow v_{opt} \leq 0$$

Finally from $v_{opt} \geq 0$ and $v_{opt} \leq 0$, we get $v_{opt} = 0$.

4. You manage the operations of a tea company. Your company purchases 5 raw ingredients: green tea, black tea, jasmine, citrus peel, spices, and then blends them in various ratios to produce 3 different blended teas which are then sold.

- (a) The proportion of raw ingredients in the 3 blended teas are, by weight:

Table 1: Blend compositions (proportion by weight)

	Green tea	Black Tea	Jasmine	Citrus peel	Spices
Blend 1	70%	0%	30%	0%	0%
Blend 2	0%	80%	0%	20%	0%
Blend 3	0%	80%	10%	0%	10%

- (b) The raw ingredients can be purchased every month. The cost (per lb) of purchasing the raw ingredient in the next 6 months are:

Table 2: Costs of raw ingredients (per lb)

	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Green tea	\$2	\$3	\$5	\$4	\$3	\$2
Black Tea	\$3	\$2	\$3	\$3	\$4	\$4
Jasmine	\$1	\$1	\$1	\$2	\$1	\$1
Citrus peel	\$1	\$1	\$1	\$1	\$1	\$1
Spices	\$4	\$5	\$6	\$7	\$5	\$4

- (c) The production of the three blends can be done every month. The cost of production is \$2 per lb, independent of blend type or month of production. The selling price (per lb) of the three blends are: Blend 1: \$20, Blend 2: \$19, Blend 3: \$23. These selling prices are the same each month.
- (d) The demand for blended tea of all three types put together for the next six months is: 200, 250, 300, 200, 250, 200 lbs. These demands must be met exactly. Also based on market survey, at least 30% of the tea sold each month must be Blend 1, at least 30% of tea sold each month must be Blend 2, and at least 20% of tea sold each month must be Blend 3. [Example: For the first month, 60 lbs of Blend 1, 60 lbs of Blend 2, and 80 lbs of Blend 3 is feasible, since then it satisfies exactly the total tea demand of month 1 ($60 + 60 + 80 = 200$ lbs), and percentage of Blend 1 is 30% ($\geq 30\%$), percentage of Blend 2 is 30% ($\geq 30\%$), and percentage of Blend 3 is 40% ($\geq 20\%$).]
- (e) Note that an inventory of both raw ingredients and the three blends can be maintained. There is no cost of storing inventory. Initial inventories of both raw ingredients and tea blends are zero. There should be no inventories of raw ingredients and tea blends at the end of six month.

Write an LP formulation to maximize profit [selling price - (production cost + purchasing cost)], that determines the amount of each of the raw ingredients to be purchased per month and the amount of different blends sold every month.

Solution:

Let's introduce following notation:

- $\mathcal{M} = \{1, 2, 3, 4, 5, 6\}$ set of months
- $\mathcal{R} = \{\text{green tea, black tea, jasmine, citrus peel, spices}\}$ set of raw ingredients
- $\mathcal{B} = \{1, 2, 3\}$ set of blends
- c_{ij} , $i \in \mathcal{M}, j \in \mathcal{R}$ price of raw ingredient i in month j from Table 2
- p_{ij} , $i \in \mathcal{B}, j \in \mathcal{R}$ percentage of raw ingredient i in blend j from Table 1
- $c = 2$ cost of production of (any) tea blend per pound
- π_i , $i \in \mathcal{B}$ selling price per pound of blend i , meaning $\pi_1 = 20$, $\pi_2 = 19$, $\pi_3 = 23$.
- D_i , $i \in \mathcal{M}$ demand for blended tea (all three types together) in month i , meaning $D_1 = 200, D_2 = 250, D_3 = 300, D_4 = 200, D_5 = 250, D_6 = 200$

Decision variables:

- r_{ij} , $i \in \mathcal{M}, j \in \mathcal{R}$: amount, in pounds, of raw ingredient i bought in month j
- R_{ij} $i \in \mathcal{M}, j \in \mathcal{R}$: inventory, in pounds, of raw ingredient i at the end of month j
- b_{ij} , $i \in \mathcal{M}, j \in \mathcal{B}$: amount, in pounds, of blend i produced in month j
- B_{ij} $i \in \mathcal{M}, j \in \mathcal{B}$: inventory, in pounds, of blend i at the end of month j
- s_{ij} , $i \in \mathcal{M}, j \in \mathcal{B}$: amount, in pounds, of blend i sold in month j

Goal is to maximize profit. To obtain function for profit we need following:

- **Revenue:** $\sum_{i=1}^6 \sum_{j=1}^3 \pi_i s_{ij}$ is revenue obtained by selling tea blends
- **Production cost:** $2 \sum_{i=1}^6 \sum_{j=1}^3 b_{ij}$ is cost of producing tea blends
- **Purchasing cost:** $\sum_{i=1}^6 \sum_{j \in \mathcal{R}} c_{ij} r_{ij}$

We obtain objective function as difference between revenue and costs.

Objective function:

$$\sum_{i=1}^6 \sum_{j=1}^3 \pi_j s_{ij} - \left(2 \sum_{i=1}^6 \sum_{j=1}^3 b_{ij} + \sum_{i=1}^6 \sum_{j \in \mathcal{R}} c_{ij} r_{ij} \right)$$

Constraints:

- **Monthly demands must be met:**

$$\sum_{j=1}^3 s_{ij} = D_i, \quad \forall j \in \mathcal{M}$$

- **At least 30% of the tea sold each month must be Blend 1:**

$$0.3D_i \leq s_{i,1}, \forall i \in \mathcal{M}$$

- **At least 30% of the tea sold each month must be Blend 2:**

$$0.3D_i \leq s_{i,2}, \forall i \in \mathcal{M}$$

- **At least 20% of the tea sold each month must be Blend 3:**

$$0.2D_i \leq s_{i,3}, \forall i \in \mathcal{M}$$

Next we need to write constraints to ensure inventory balances of raw ingredients and tea blends at the end of each month. For raw ingredients that means that for each raw ingredient, at the end of each month, amount of the raw ingredient in the inventory should be equal to the amount of the raw ingredient in the inventory at the end of previous month plus amount of the raw ingredient bought in the current month minus amount of the raw ingredient used to produce tea blends during the current month. Note that inventory of raw ingredients before month 1 was zero and inventory of the raw ingredients at the end of month 6 should be zero.

- **Inventory balance for the raw ingredients at the end of the first month:**

$$R_{1,j} = r_{1,j} - \sum_{i=1}^3 p_{ij} b_{1,i}, \forall j \in \mathcal{R}$$

- **Inventory balance for the raw ingredients at the end of the months 2, ..., 6:**

$$R_{ij} = R_{i-1,j} + r_{ij} - \sum_{k=1}^3 p_{kj} b_{ik}, \forall i \in \mathcal{M} \setminus \{1, 6\}, \forall j \in \mathcal{R}$$

- **Inventory balance for the row ingredients at the end of the last month must be zero:**

$$R_{6,j} = R_{5,j} + r_{6,j} - \sum_{i=1}^3 p_{ij} b_{6,i}, \forall j \in \mathcal{R}$$

For blends that means that for each blend, at the end of each month, amount of the blend in the inventory should be equal to the amount of the blend in the inventory at the end of previous month plus amount of the blend produced during the current month minus amount of the blend sold during the current month. Note that inventory of the blends before month 1 was zero and inventory of the blends at the end of month 6 should be zero.

- **Inventory balance for the tea blends at the end of the first month:**

$$B_{1,j} = b_{1,j} - s_{1,j}, \forall j \in \mathcal{B}$$

- **Inventory balance for the tea blends at the end of the months 2, ..., 6:**

$$B_{ij} = B_{i-1,j} + b_{ij} - s_{ij}, \forall i \in \mathcal{M} \setminus \{1\}, \forall j \in \mathcal{B}$$

- **Inventory balance for the tea blends at the end of the last month must be zero:**

$$0 = B_{5,j} + b_{6,j} - s_{ij}, \quad \forall j \in \mathcal{B}$$

Finally, all of the variables should be nonnegative

- Variable nonnegativity:

$$r_{ij}, R_{ij} \geq 0, \quad i \in \mathcal{M}, j \in \mathcal{R}$$

$$b_{ij}, B_{ij}, s_{ij} \geq 0, \quad i \in \mathcal{M}, j \in \mathcal{B}$$

Finally we can put it all together:

$$\begin{aligned}
\max \quad & \sum_{i=1}^6 \sum_{j=1}^3 \pi_j s_{ij} - \left(2 \sum_{i=1}^6 \sum_{j=1}^3 b_{ij} + \sum_{i=1}^6 \sum_{j \in \mathcal{R}} c_{ij} r_{ij} \right) \\
\text{s.t.} \quad & \sum_{j=1}^3 s_{ij} = D_i, \quad \forall i \in \mathcal{M} \\
& 0.3D_i \leq s_{i,1}, \quad \forall i \in \mathcal{M} \\
& 0.3D_i \leq s_{i,2}, \quad \forall i \in \mathcal{M} \\
& 0.2D_i \leq s_{i,3}, \quad \forall i \in \mathcal{M} \\
& R_{1,j} = r_{1,j} - \sum_{i=1}^3 p_{ij} b_{1,i}, \quad \forall j \in \mathcal{R} \\
& R_{ij} = R_{i-1,j} + r_{ij} - \sum_{k=1}^3 p_{kj} b_{ik}, \quad \forall i \in \mathcal{M} \setminus \{1, 6\}, \quad \forall j \in \mathcal{R} \\
& 0 = R_{5,j} + r_{6,j} - \sum_{i=1}^3 p_{ij} b_{6,i}, \quad \forall j \in \mathcal{R} \\
& B_{1,j} = b_{1,j} - s_{1,j}, \quad \forall j \in \mathcal{B} \\
& B_{ij} = B_{i-1,j} + b_{ij} - s_{ij}, \quad \forall i \in \mathcal{M} \setminus \{1, 6\}, \quad \forall j \in \mathcal{B} \\
& 0 = B_{5,j} + b_{6,j} - s_{6,j}, \quad \forall j \in \mathcal{B} \\
& r_{ij}, R_{ij} \geq 0, \quad i \in \mathcal{M}, j \in \mathcal{R} \\
& b_{ij}, B_{ij}, s_{ij} \geq 0, \quad i \in \mathcal{M}, j \in \mathcal{B}
\end{aligned}$$

Note: Problem only asked for the formulation. If you want to check your solution, you can implement it in CVXPY and obtain solution. Optimal value should be \$23,702.